$h(x)=\sqrt{x+45}-\sqrt{x+5}$ 求 $[\sum_{i=1}^{n}x_i]$

## g(x) 性质

$g^{'}(x)=\frac{1}{2\sqrt{x+45}}-\frac{1}{2\sqrt{x+5}}$ $g^{"}(x)=-\frac{1}{4(x+45)^{3/2}}+\frac{1}{4(x+5)^{3/2}}>0$
$x>0$ 时，g(x)↑g(x) 下凸

## 背景

由第一问得 $3.9<x_{2n}<4.0, 4.0<x_{2n=1}<4.1$ 发现 $|x_n-4|\downarrow$ 且 $\to 0$

## 构造直线

$g^{'}(4)=-\frac{2}{21}$
构造 3 条直线：

$$a(x)=-\frac{2}{21}(x-4)+4$$

$$a(x)=-\frac{1}{4}(x-4)+4$$

$$a(x)=-k(x-4)+4$$

易证：

$$0<x<4, a(x)<g(x)<b(x)(01)$$

$$4<x<5, a(x)<g(x)<c(x)(02)$$

## 放缩

$\because 0<x_{2k}<4$ 且 (01)
$\therefore a(x_{2k})<g(x_{2k})<b(x_{2k})$
$\because a(x)\downarrow$
$\therefore a(x_{2k})>a(4)=4$
$\because g(x_{2k})=x_{2k+1}$
$\therefore 4<a(x_{2k})<x_{2k+1}<b(x_{2k})$
即

$$-\frac{2}{21}(x_{2k}-4)+4<x_{2k+1}<-\frac{1}{4}(x_{2k}-4)+4(1)$$

令 $k=1$

$$-\frac{2}{21}(x_2-4)<x_3-4<-\frac{1}{4}(x_2-4)(7)$$

$\because 0<X<4, c(x)>g(x)>a(x)$
$\therefore c(a(x_{2k}))>g(a(x_{2k})), g(x_{2k+1})>a(x_{2k=1})$
$\because c, g, a\downarrow$
$\therefore g(a(x_{2k}))>g(x_{2k+1}), a(x_{2k+1})>a(b(x_{2k}))$
$\therefore c(a(x_{2k}))>g(x_{2k+1})>a(b(x_{2k}))$
即

$$-2/21[-1/4(x_{2k}-4)]+4<x_{2k+2}<-k[-2/21(x_{2k-4})]+4$$

$$\frac{1}{42}(x_{2k}-4)<x_{2k+2}-4<\frac{2}{21}k(x_{2k-4})(2)$$

$$\frac{1}{1-1/42}(x_2-4)<\frac{1-(\frac{1}{42})^n}{1-\frac{1}{42}}(x_2-4)<\sum_{i=1}^{n}x_{2k}-4<\frac{1-(\frac{2k}{21})^n}{1-\frac{2k}{21}}(x_2-4)<(x_2-4)$$

即

$$\frac{1}{1-1/42}(x_2-4)<\sum_{i=1}^{n}x_{2k}-4<(x_2-4)(3)$$

类似的，可得

$$-\frac{2}{21}(x_{2k-1}-4)+4<x_{2k}<-k(x_{2k-1}-4)+4()$$

令 $k=1$

$$-\frac{2}{21}(x_1-4)<x_2-4<-k(x_1-4)(9)$$

$$\frac{1}{1-1/42}(x_3-4)>\frac{1-(\frac{1}{42})^n}{1-\frac{1}{42}}(x_3-4)>\sum_{i=1}^{n}x_{2k+1}-4>\frac{1-(\frac{2k}{21})^n}{1-\frac{2k}{21}}(x_3-4)>(x_3-4)$$

即

$$\frac{1}{1-1/42}(x_3-4)>\sum_{i=1}^{n}x_{2k+1}-4>(x_3-4)(6)$$

由 (3)(6)

$$(x_3-4)+\frac{1}{1-1/42}(x_2-4)<\sum_{i=2}^{n}x_i-4<\frac{1}{1-1/42}(x_3-4)+(x_2-4)$$

由 (7)

$$(-\frac{2}{21}+\frac{1}{1-\frac{1}{42}})(x_2-4)<\sum_{i=2}^{n}x_i-4<(\frac{1}{1-\frac{1}{42}}(-\frac{1}{4})+1)(x_2-4)$$

$\because 0<-\frac{2}{21}+\frac{1}{1-\frac{1}{42}}<1$ 且 $(x_2-4)<0$
$\therefore (-\frac{2}{21}+\frac{1}{1-\frac{1}{42}})(x_2-4)>(x_2-4)$
$\because \frac{1}{1-\frac{1}{42}}(-\frac{1}{4})+1>0$ 且 $(x_2-4)<0$
$\therefore (\frac{1}{1-\frac{1}{42}}(-\frac{1}{4})+1)(x_2-4)<0$
得

$$(x_2-4)<\sum_{i=2}^{n}x_i-4<0$$

由 (9)

$$-\frac{2}{21}(x_1-4)<\sum_{i=2}^{n}x_i-4<0$$

即

$$-\frac{2}{21}<\sum_{i=2}^{n}x_i-4<0$$

$\therefore$

$$-\frac{2}{21}<(\sum_{i=2}^{n}x_i)-4(n-1)<0$$

$$-\frac{2}{21}+4(n-1)<(\sum_{i=2}^{n}x_i)<4(n-1)$$

$$1-\frac{2}{21}+4n<(\sum_{i=2}^{n}x_i)+5<4n+1$$

$$1-\frac{2}{21}+4n<\sum_{i=1}^{n}x_i<4n+1$$

$\therefore$

$$[\sum_{i=1}^{n}x_i]=4n$$